

Class 05: Data Visualization with GGLOT

Hubert Phan (PID: A17663330)

Table of contents

Background	1
Common Plot Types	4
Creating Scatter Plots	5
Gene Expression Plot	7
Going Further	11

Background

There are two ways to create visualizations in R: using base R functions or using the ggplot2 package. Data visualizations created with ggplot2 are often more aesthetically pleasing and easier to customize than those created with base R functions.

This is an example of making a scatter plot of the cars dataset using base R functions and ggplot2:

```
head(cars,3)
```

```
      speed dist
1         4    2
2         4   10
3         7    4
```

Base R:

```
plot(cars)
```

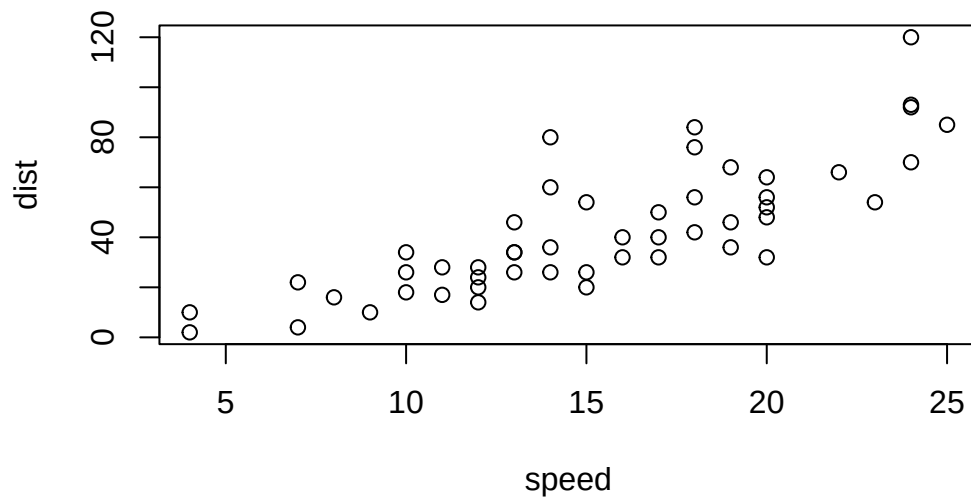


Figure 1: Base R plot of the cars dataset

ggplot2:

```
#install.packages("ggplot2")  
library(ggplot2)  
ggplot(cars)
```

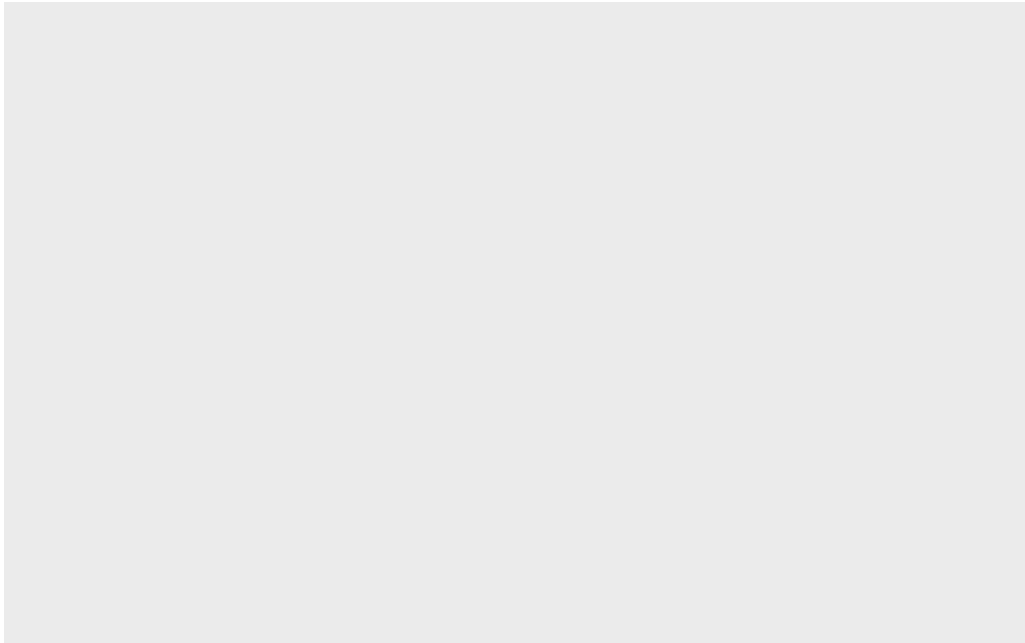


Figure 2: example ggplot2 without setting dataset

Every ggplot needs atleast 3 things:

- A dataset
- Aesthetic mappings (aes)
- Geometric objects (geoms)

```
library(ggplot2)
ggplot(data = cars, aes(x = speed, y = dist)) +
  geom_point()
```

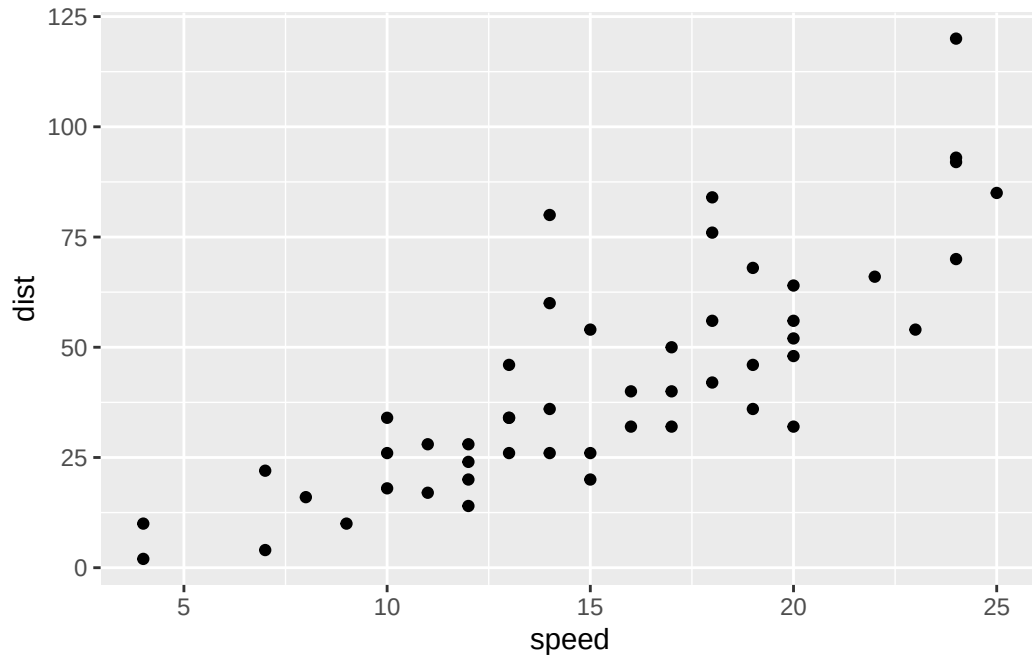


Figure 3: ggplot2 scatter plot example

Q1. For which phases is data visualization important in our scientific workflows?

Ans: Communication of Results, Exploratory Data Analysis (EDA), and Detection of outliers.

Q2. True or False? The ggplot2 package comes already installed with R?

Ans: False

Common Plot Types

Q. Which plot types are typically NOT used to compare distributions of numeric variables?

Ans: Network graphs

Q. Which statement about data visualization with ggplot2 is incorrect?

Ans: ggplot2 is the only way to create plots in R

Creating Scatter Plots

Q. Which geometric layer should be used to create scatter plots in ggplot2?

Ans: `geom_point()`

Note: These next questions refer to the ggplot scatter plot created from the cars dataset above.

Q. In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the `geom_smooth()` function?

```
ggplot(data = cars, aes(x = speed, y = dist)) +  
  geom_point() +  
  geom_smooth()
```

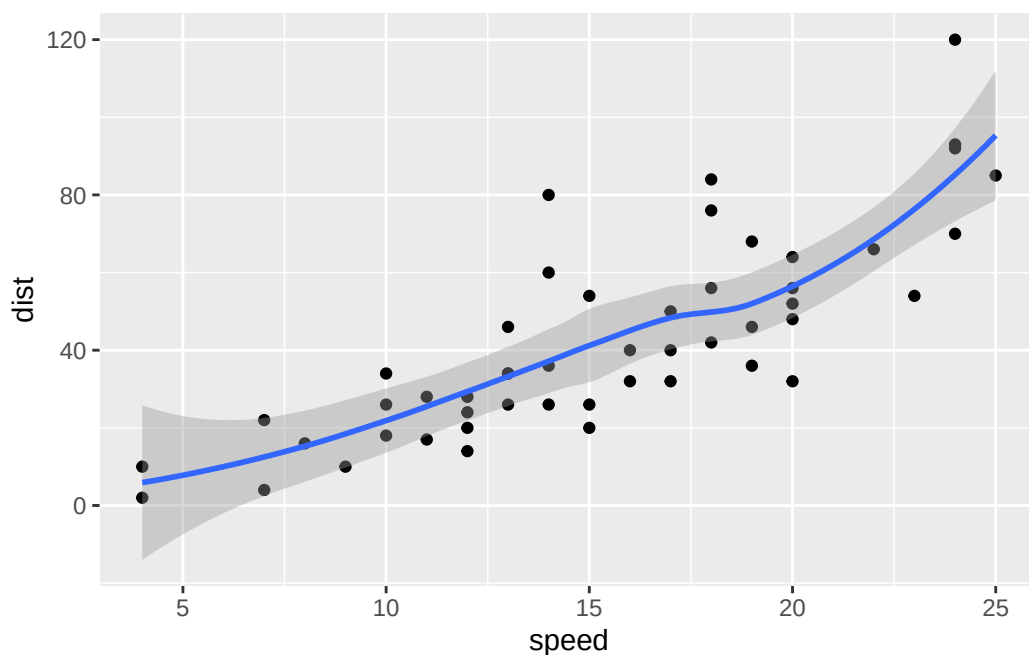


Figure 4: ggplot2 scatter plot with trend line

Q. Argue with `geom_smooth()` to add a straight line from a linear model without the shaded standard error region?

```
ggplot(data = cars, aes(x = speed, y = dist)) +  
  geom_point() +  
  geom_smooth(method = "lm", se = FALSE)
```

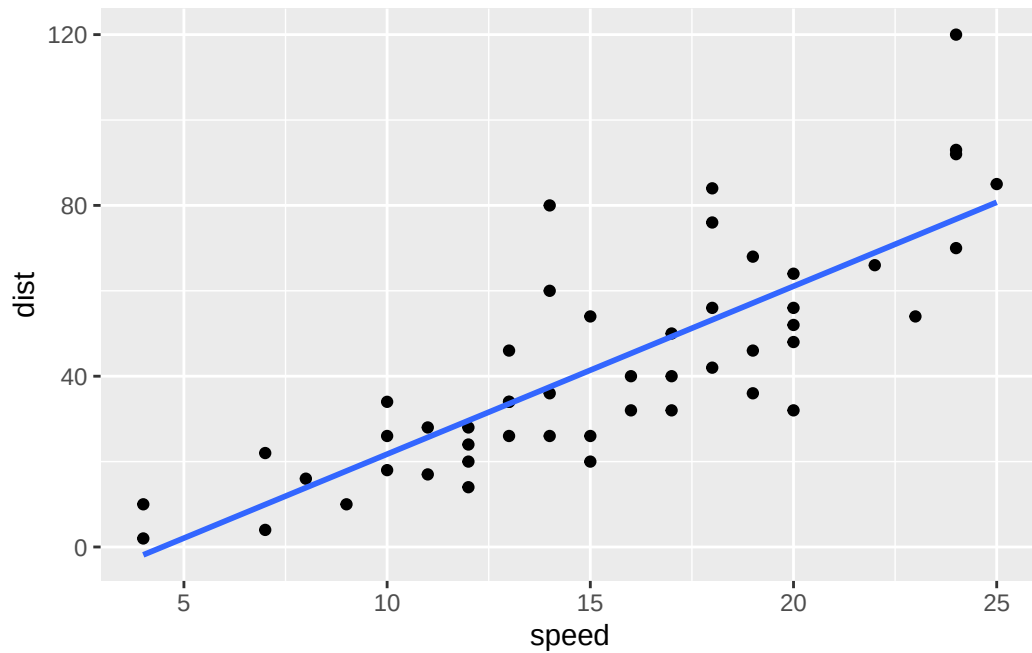


Figure 5: ggplot2 scatter plot with linear trend line

Q. Can you finish this plot by adding various label annotations with the `labs()` function and changing the plot look to a more conservative “black & white” theme by adding the `theme_bw()` function?

```
ggplot(data = cars, aes(x = speed, y = dist)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Stopping Distance vs. Speed of Cars",
       x = "Speed (mph)",
       y = "Stopping Distance (ft)") +
  theme_bw()
```

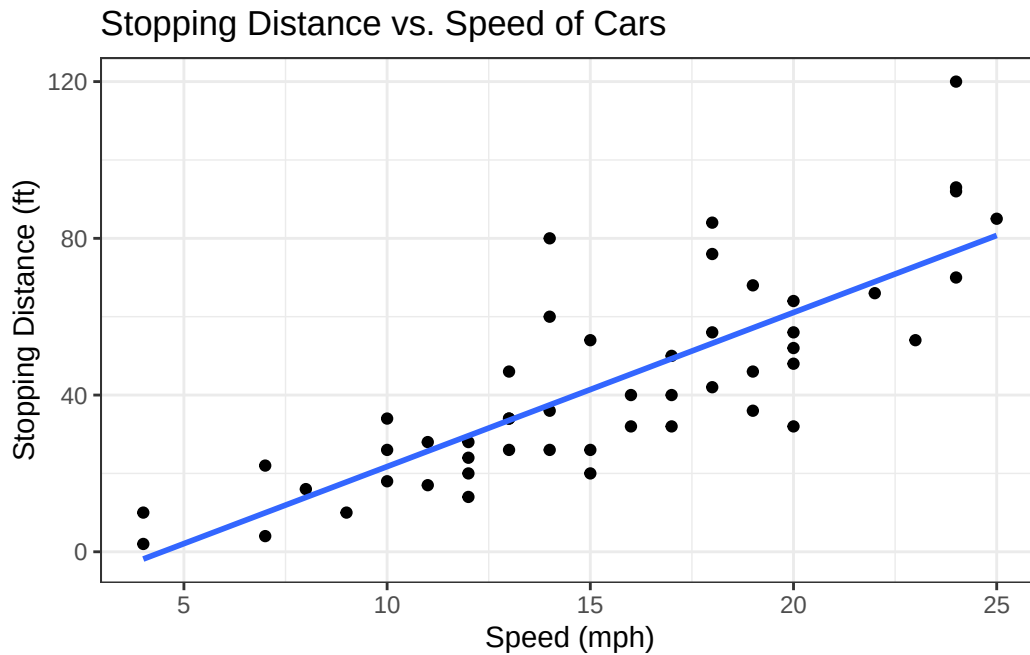


Figure 6: ggplot2 scatter plot with labels and black & white theme

Gene Expression Plot

Read the expression data from the class website:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Q. Use the `nrow()` function to find out how many genes are in this dataset. What is your answer?

```
nrow(genes)
```

```
[1] 5196
```

Q. Use the `colnames()` function and the `ncol()` function on the `genes` data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

```
colnames(genes)
```

```
[1] "Gene"          "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

Q. Use the `table()` function on the `State` column of this data.frame to find out how many 'up' regulated genes there are. What is your answer?

```
table(genes$State)
```

down	unchanging	up
72	4997	127

Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

```
total_upreg <- table(genes$State)["up"]
total_genes <- nrow(genes)
fraction_upreg <- round(total_upreg / total_genes, 2)
fraction_upreg
```

```
up
0.02
```

First Expression Scatter Plot

```
ggplot(genes, aes(x = Condition1, y = Condition2)) +  
  geom_point()
```

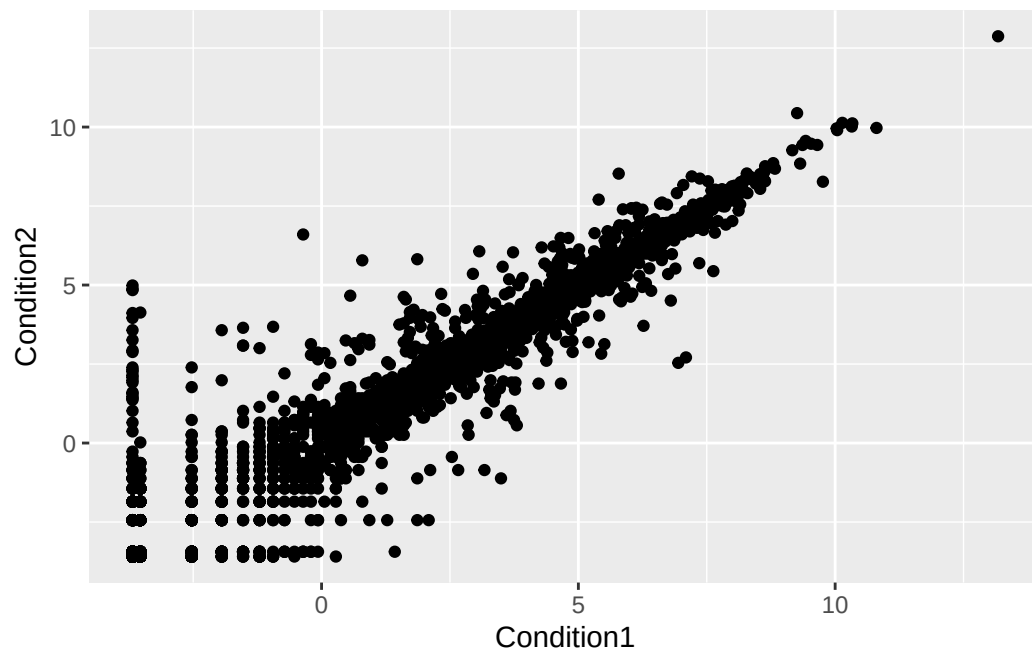


Figure 7: Gene expression scatter plot

Second Expression Scatter Plot

```
ggplot(genes, aes(x = Condition1, y = Condition2, color = State)) +  
  geom_point()
```

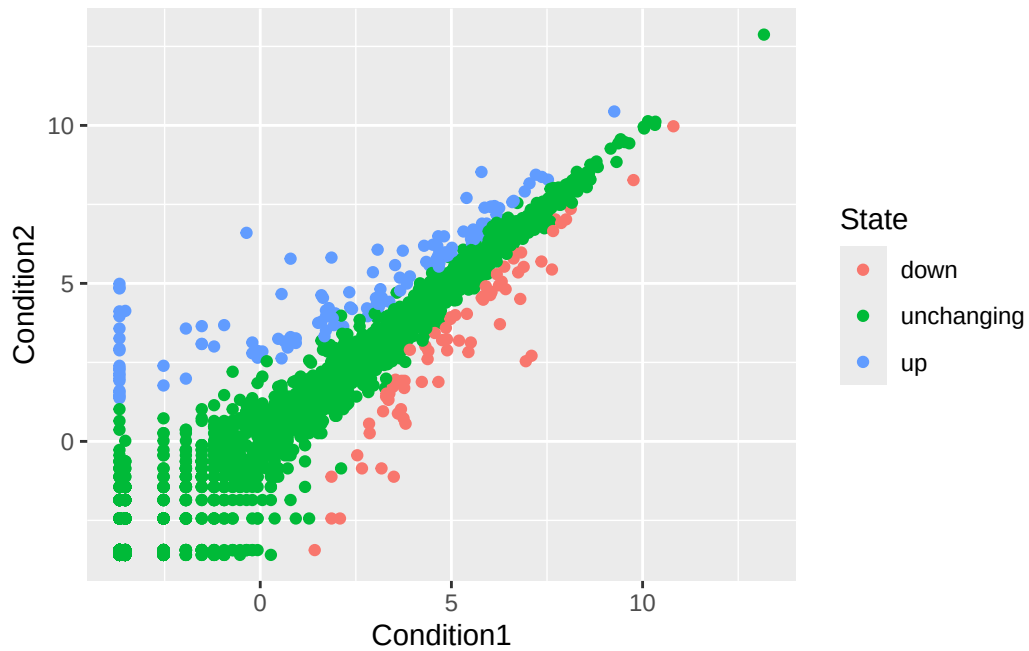


Figure 8: Gene expression scatter plot colored by State

Third Expression Scatter Plot

```
ggplot(genes, aes(x = Condition1, y = Condition2, color = State)) +
  geom_point() +
  scale_color_manual(values = c("blue", "gray", "red")) +
  labs(title = "Gene Expression Changes Upon Drug Treatment",
       x = "Control (no drug)",
       y = "Drug Treatment") +
  theme_bw()
```

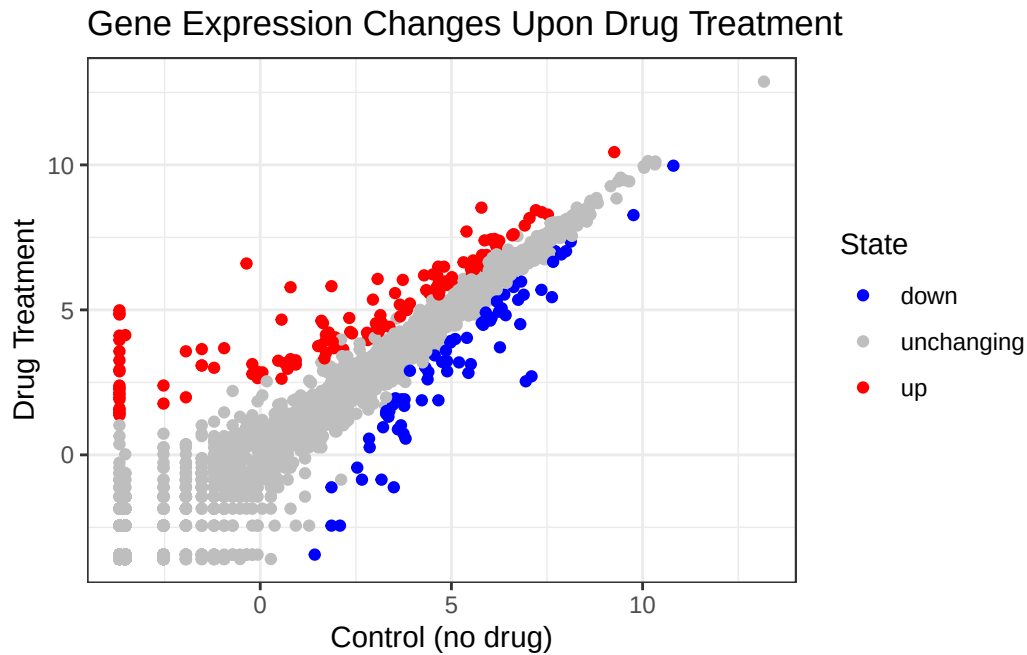


Figure 9: Gene expression scatter plot with labels and theme

Going Further

Read the gapminder dataset:

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"
gapminder <- read.delim(url)
head(gapminder)
```

	country	continent	year	lifeExp	pop	gdpPercap
1	Afghanistan	Asia	1952	28.801	8425333	779.4453
2	Afghanistan	Asia	1957	30.332	9240934	820.8530
3	Afghanistan	Asia	1962	31.997	10267083	853.1007
4	Afghanistan	Asia	1967	34.020	11537966	836.1971
5	Afghanistan	Asia	1972	36.088	13079460	739.9811
6	Afghanistan	Asia	1977	38.438	14880372	786.1134

Use dplyr to filter for the year 2007:

```
library(dplyr)
gapminder2007 <- gapminder %>% filter(year == 2007)
head(gapminder2007)
```

	country	continent	year	lifeExp	pop	gdpPerCap
1	Afghanistan	Asia	2007	43.828	31889923	974.5803
2	Albania	Europe	2007	76.423	3600523	5937.0295
3	Algeria	Africa	2007	72.301	33333216	6223.3675
4	Angola	Africa	2007	42.731	12420476	4797.2313
5	Argentina	Americas	2007	75.320	40301927	12779.3796
6	Australia	Oceania	2007	81.235	20434176	34435.3674

Q. Complete the code below to produce a first basic scatter plot of this gapminder_2007 dataset:

```
ggplot(gapminder2007, aes(x = gdpPerCap, y = lifeExp)) +
  geom_point()
```

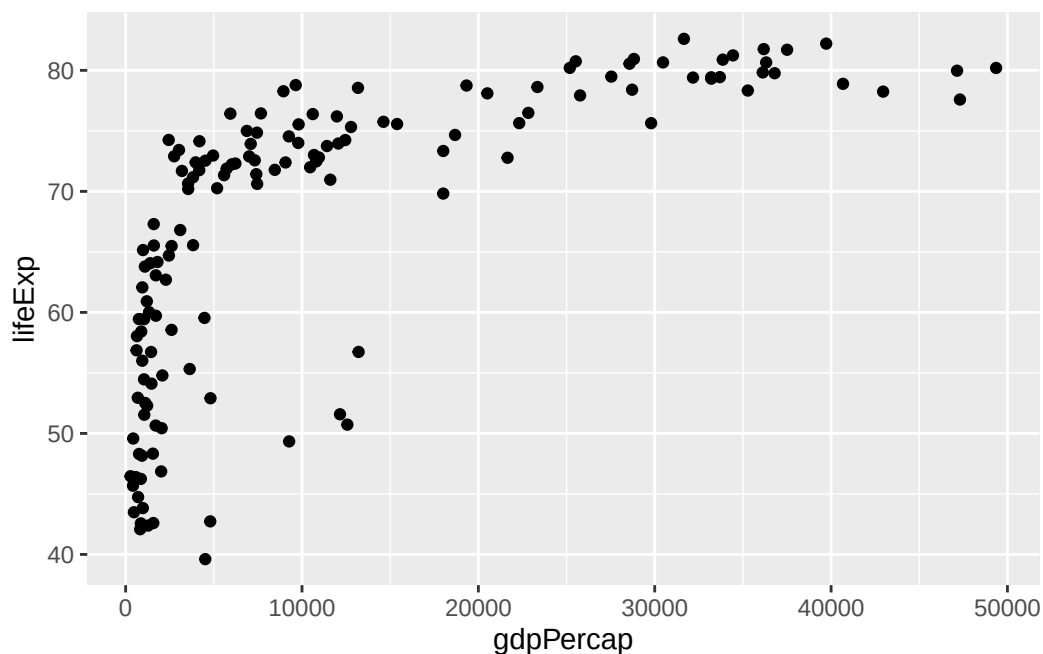


Figure 10: Gapminder 2007 scatter plot of GDP per capita vs Life Expectancy

Q. Can you adapt the code you have learned thus far to reproduce our gapminder scatter plot for the year 1957? What do you notice about this plot is it easy to compare with the one for 2007?

```
gapminder_1957 <- gapminder %>% filter(year == 1957)
ggplot(gapminder_1957, aes(x = gdpPercap, y = lifeExp)) +
  geom_point(aes(color = continent, size = pop), alpha = 0.7) +
  scale_size_area(max_size = 15)
```

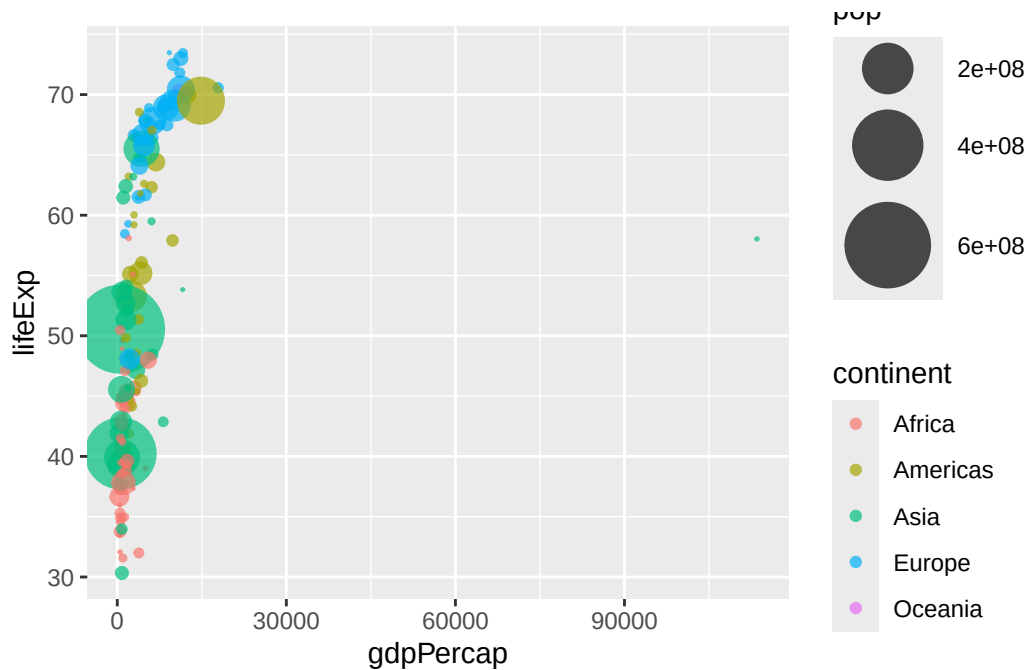


Figure 11: Gapminder scatter plot for 1957

Q. Do the same steps above but include 1957 and 2007 in your input dataset for `ggplot()`. You should now include the layer `facet_wrap(~year)` to produce the following plot:

```
gapminder_5707 <- gapminder %>% filter(year == 1957 | year == 2007)
ggplot(gapminder_5707, aes(x = gdpPercap, y = lifeExp)) +
  geom_point(aes(color = continent, size = pop), alpha = 0.7) +
  scale_size_area(max_size = 8) +
  facet_wrap(~year)
```

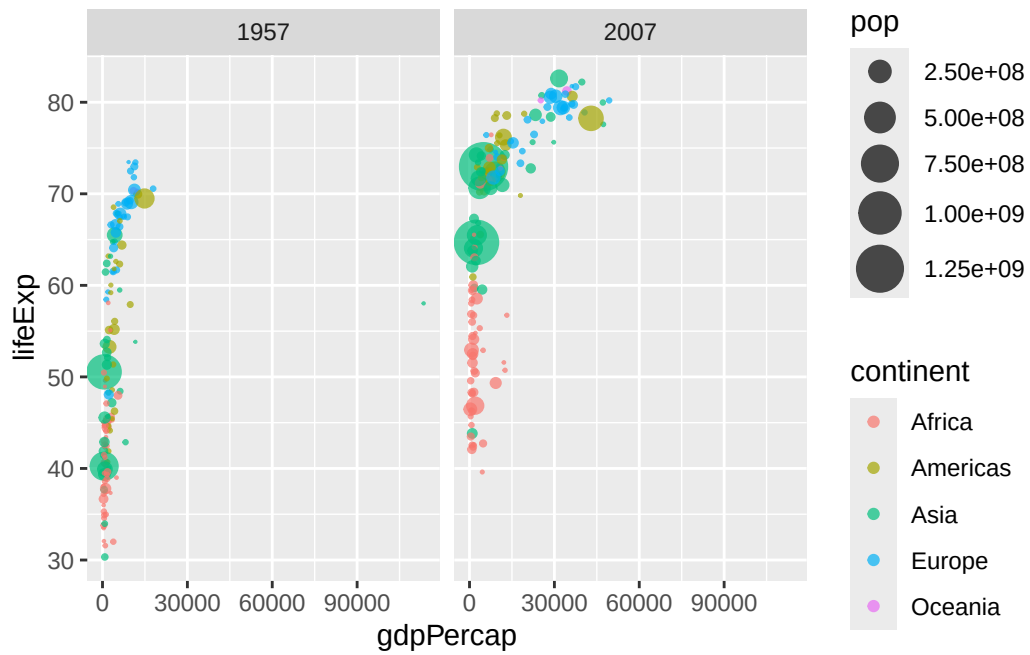


Figure 12: Gapminder scatter plot faceted by year